CAPSTONE 3 — BANKING TRANSACTION, RISK & EXPOSURE ANALYTICS

Business Scenario: Banking Transaction, Risk & Exposure Analytics Capstone

End-to-End BI Engineering Capstone (40 Hours)

Organization Background

A mid-to-large commercial bank operates across multiple regions through a wide branch network, serving both retail and small-to-mid corporate customers. The bank processes thousands of financial transactions daily across savings, current, and credit-linked accounts.

In an increasingly regulated environment, the bank faces growing pressure from regulators and internal risk teams to demonstrate stronger control, transparency, and explainability in transaction monitoring and risk exposure reporting.

Business Problem

While transaction volumes continue to rise, senior management and compliance teams face key challenges:

- Limited visibility into high-risk and anomalous transactions

- Difficulty linking transactions, accounts, customers, and branches into a unified risk view

- Inconsistent reporting of credit vs debit exposure

- Reactive fraud and risk monitoring driven by static reports

- Regulatory concerns around data quality, audit trails, and KPI explainability

Different departments (Risk, Compliance, Operations, Finance) often report conflicting metrics due to inconsistent definitions and lack of a centralized analytical model.

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Objective of the Capstone

The objective of this capstone is to design and deliver a regulator-safe Banking Transaction, Risk & Exposure Analytics

BI solution that:

- Establishes a single, auditable version of truth for transaction and risk KPIs
- Enables transparent identification of high-risk transactions and accounts
- Quantifies financial exposure at account, branch, and bank levels
- Supports proactive risk monitoring and executive decision-making

Scope of Analysis

The analytics solution will support:

- End-to-end transaction monitoring
- Credit vs debit behavior analysis
- High-value and high-risk transaction identification
- Account-level and branch-level exposure analysis
- Risk flag aggregation and anomaly detection
- Time-based trend analysis for transactions and risks

Real-World Data & Regulatory Challenges

The datasets intentionally simulate real banking system complexities:

- Mixed debit and credit sign conventions
- Missing or inconsistent balances
- Duplicate transaction and risk records
- Orphan risk flags not linked to transactions

- Extreme transaction values and negative balances
- Injected data quality issues reflecting upstream system failures

All transformations and logic must remain fully explainable, auditable, and compliant with regulatory expectations, using BI tools only.

Expected Business Outcomes

Upon completion, stakeholders expect:

- Accurate, defensible transaction and risk KPIs
- Clear identification of high-risk customers, accounts, and branches
- Improved understanding of financial exposure and risk concentration
- Strong audit trails supporting regulatory reviews
- A scalable BI framework reusable for future banking analytics initiatives

Success Criteria

The solution will be considered successful when:

- Transaction, risk, and exposure KPIs reconcile across Power BI, Tableau, and Qlik Sense
- Risk logic and thresholds are fully explainable and parameter-driven
- Dashboards support drill-down from bank-level to transaction-level detail
- Data quality and risk flags are transparent, not hidden
- The solution withstands executive, regulatory, and interview-level scrutiny

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CAPSTONE CONSTRAINTS (MANDATORY TO BE FOLLOWED)

- **Pure BI Engineering only**
    - Power BI
    - Tableau
    - Qlik Sense
- ❌ No SQL engines
- ❌ No Python
- ❌ No external ETL / cloud tools
- All transformations must be done using:
    - Power Query
    - Tableau Prep
    - Qlik Load Script
- Raw datasets must never be overwritten
- All logic must be **explainable, auditable, regulator-safe**

DATASETS Overview: (BANKING)

Banking_Capstone3/

├── transactions.csv (~10,000 rows)

├── accounts.csv (~2,500 rows)

├── customers.csv (~2,000 rows)

├── branches.csv (~120 rows)

├── risk_flags.csv (~15,000 rows)

├── calendar.csv (120 days)

└── data_quality_injections.csv

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🟦 DAYS 1–2.5 — BI DATA ENGINEERING (20 HOURS)

🟦 DAY 1 — DATA UNDERSTANDING & PROFILING (8 HOURS)

Objective

Understand **banking transaction behavior, risk signals, and regulatory risks** before any transformation.

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1. Business KPI & Risk Alignment (1.5 hrs)

Generic Tasks

- Confirm banking KPIs:
  - Total Transactions
  - Transaction Volume
  - Credit vs Debit Split
  - Average Transaction Value
  - High-Risk Transaction Count
  - Fraud-Suspected Transactions
  - Account Balance Exposure
  - Branch Risk Distribution
- Define **risk interpretation rules**:
  - High-value threshold
  - Dormant account logic
  - Multi-risk aggregation
- Map KPIs → required fields

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2. Raw Data Ingestion (2 hrs)

Generic Tasks

- Load datasets **as-is**
- Preserve RAW layer
- No cleansing or filtering

Datasets

- Transactions
- Accounts
- Customers
- Branches
- Risk Flags
- Calendar

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3. Data Profiling & Diagnostics (3 hrs)

Generic Tasks

- Identify:
    - Mixed debit/credit signs
    - Null or missing balances
    - Mixed datetime formats
    - Duplicate transaction IDs
    - Orphan risk flags
- Profile:
    - Transaction volume skew
    - Account and branch cardinality
- Document regulatory risks

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4. Grain & Join Validation (1.5 hrs)

Generic Tasks

- Confirm fact grain:
  - **1 row = 1 financial transaction**
- Validate joins:
  - Customer → Account → Transaction
  - Transaction → Risk Flags (1:M)
- Identify need for UNKNOWN members

✅ **DAY 1 DONE IF**

✓ Raw data preserved

✓ Risk interpretation documented

✓ Join & grain risks identified

🟦 DAY 2 — DATA CLEANSING & FINANCIAL STANDARDIZATION (8 HOURS)

Objective

Create **clean, regulator-safe banking datasets** with consistent financial logic.

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5. Date & Amount Standardization (2 hrs)

Generic Tasks

- Normalize:
  - Transaction timestamps
  - Risk timestamps
- Remove currency symbols & commas
- Convert to decimals

- Flag parse failures

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6. Deduplication Logic (2 hrs)

Generic Tasks

- Deduplicate:
    - Transactions → keep latest by created_ts
    - Risk flags → keep latest per risk_flag_id
- Preserve duplicate flags for audit

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7. Missing & Invalid Value Handling (1.5 hrs)

Generic Tasks

- Handle missing:
    - account_id
    - customer_id
    - branch_id
- Implement UNKNOWN strategy
- Validate closed-account activity

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8. Anomaly & Risk Engineering (1.5 hrs)

Generic Tasks

- Flag:
    - Debit/Credit sign mismatches
    - Extreme transaction values
    - Extreme negative balances
    - Risk flags earlier than transactions
- Parameterize thresholds

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9. Audit & Reusability Engineering (1 hr)

Generic Tasks

- Add:
  - DQ flags
  - Risk flags
  - Lineage columns
- Modularize logic
- Parameterize risk thresholds

✅ **DAY 2 DONE IF**

✓ Clean transactions & risk flags

✓ Financial logic standardized

✓ Audit flags present

🟦 DAY 2.5 — MODELING & RISK SEMANTIC READINESS (4 HOURS)

Objective

Prepare **BI-optimized star schema** for financial & risk analytics.

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10. Dimensional Modeling (2 hrs)

Generic Tasks

- Build star schema:
  - FactTransactions
  - DimCustomer
  - DimAccount

  - DimBranch
  - DimDate
- Optional:
  - FactRiskFlags (or rolled-up risk attributes)
- Add UNKNOWN dimension rows

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11. Risk Semantic Readiness (2 hrs)

Generic Tasks

- Prepare:
  - High-risk indicators
  - Risk count per transaction
  - Exposure measures
- Hide technical fields
- Validate aggregation safety

✅ **BI DATA ENGINEERING COMPLETE**

✓ Clean data

✓ Stable model

✓ Risk-ready semantics

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🟩 DAYS 3–5 — BI REPORTING & ANALYTICAL ENGINEERING (20 HOURS)

🟩 DAY 3 — KPI SEMANTICS & RISK LOGIC (8 HOURS)

Objective

Implement **correct, regulator-defensible banking KPIs**.

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12. KPI Measure Engineering (3 hrs)

Generic Tasks

- Build measures:
  - Total Transactions
  - Total Transaction Amount
  - Credit vs Debit totals
  - High-Value Transaction Count
  - High-Risk Transaction %
- Apply inclusion/exclusion rules
- Validate vs raw data

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13. Risk Computation Logic (2 hrs)

Generic Tasks

- Implement:
  - Risk aggregation per transaction
  - Exposure per account
  - Dormant account activity
- Validate edge cases

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14. Parameter & Scenario Analysis (1.5 hrs)

Generic Tasks

- What-if risk thresholds
- High-value cut-off simulation
- Validate KPI sensitivity

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15. KPI Reconciliation Engineering (1.5 hrs)

Generic Tasks

- Build visuals showing:
    - Included vs excluded transactions
    - Risk-flagged vs clean transactions
- Cross-tool validation

✅ **DAY 3 DONE IF**

✓ Risk KPIs correct

✓ Parameters functional

✓ Reconciliation proven

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🟩 DAY 4 — ANALYTICAL DASHBOARDS & INTERACTIVITY (8 HOURS)

Objective

Enable **banking risk analysis**, not static reporting.

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16. Analytical Dashboards (3 hrs)

Generic Tasks

- Build dashboards for:
    - Executive risk overview
    - Account exposure analysis
    - Branch risk distribution
    - Transaction trend analysis

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17. Interactivity Engineering (3 hrs)

Generic Tasks

- Implement:
    - Drill-down (Bank → Branch → Account → Transaction)
    - Drillthrough to transaction detail
    - Tooltips for risk explanation

Interactivity Engineering Continued:

1. **Risk Drill-down**
    - Bank → Branch → Account → Transaction
2. **High-Value Transaction Toggle**
    - Interactive threshold for defining “high value”
3. **Risk Flag Filter Panel**
    - Toggle specific risk flags on/off (velocity, amount, geo)

4. **Account Exposure Drillthrough**
    - Drillthrough from exposure KPI → transaction history
5. **Credit vs Debit Switch**
    - Toggle entire dashboard context dynamically
6. **Time-Window Risk Comparison**
    - Compare current period vs previous risk exposure
7. **Transaction Timeline Scrubber**
    - Scroll through transaction sequences interactively
8. **Risk Explanation Tooltips**
    - Hover explains why transaction is flagged
9. **Dormant Account Activation Toggle**
    - Include/exclude dormant accounts dynamically
10. **KPI Reconciliation Interaction**
    - Click excluded transactions to see exclusion reason

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18. Performance-Driven Reporting (2 hrs)

Generic Tasks

- Measure:
    - Load time
    - Filter latency
- Optimize visuals and calculations

✅ **DAY 4 DONE IF**

✓ Dashboards analytical

✓ Interactivity correct

✓ Performance optimized

🟩 DAY 5 — BI PRODUCTIZATION & REGULATORY STORYTELLING (4 HOURS)

Objective - Deliver a **production-grade, audit-ready banking BI product**.

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19. BI Product Hardening (1.5 hrs)

Generic Tasks

- Lock KPI logic
- Remove unused visuals
- Final validation

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20. Storytelling & Demo (2.5 hrs)

Generic Tasks

- Narrative:
  - Risk problem
  - Exposure drivers
  - Mitigation actions
- Executive walkthrough
- Regulatory / interview defense Q&A

✅ FINAL DEFINITION OF DONE

✓ Interactive banking dashboards

✓ Risk KPI reconciliation proof

✓ Performance optimized

✓ Audit- & interview-ready output

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Conclusion

The Banking capstone emphasizes **accuracy, auditability, and risk sensitivity**, which are non-negotiable in financial analytics.

Learners engineered transaction-level data to correctly handle **credit/debit logic, risk flags, and regulatory edge cases**, ensuring that **exposure, volume, and risk KPIs** remained defensible under scrutiny. Through careful semantic modeling and reconciliation, the capstone demonstrated how BI Engineers enable **risk visibility without compromising data integrity**.

This project highlights the importance of **explainable analytics** in banking, where every metric must be traceable back to individual transactions and policy rules.